

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**  
**III B. TECH II SEMESTER**

|                                |                                       |                                        |
|--------------------------------|---------------------------------------|----------------------------------------|
| <b>BATCH</b><br><b>2023-27</b> | <b>MINI PROJECT</b><br><b>2025-26</b> | <b>ACADEMIC YEAR</b><br><b>2025-26</b> |
|--------------------------------|---------------------------------------|----------------------------------------|

**Batch Id** : 23MPCS-H10

**Title of the Project** : KhelQuest: AI – Powered Platform for Democratizing Sports

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**GUIDE DETAILS**

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### **Objectives**

**1. Make talent identification accessible to everyone**

To give every athlete—whether from a big city or a small village—an equal chance to showcase their skills using just a smartphone.

**2. Replace complex equipment with simple, smart technology**

To allow fitness tests like jumps and push-ups to be measured accurately through AI-based video analysis instead of costly machines or specialized setups.

**3. Ensure fair and honest evaluation**

To use AI to detect incorrect movements or cheating so that every athlete's performance is judged transparently and without bias.

**4. Provide instant, meaningful feedback**

To help athletes understand their strengths and weaknesses immediately so they can improve their performance over time.

**5. Build trust through secure and verified data**

To safely store all athlete performances in a protected system so coaches and officials can rely on authentic, tamper-proof records.

**6. Support coaches and sports authorities with insights**

To offer dashboards and analytics that help experts track talent, compare performance, and make better selection decisions.

**7. Work smoothly even in rural areas**

To design the platform so it works offline or with low data usage, making it practical for remote regions with limited connectivity.

**8. Empower the nation's sports ecosystem**

To contribute to India's vision of discovering young talent early, nurturing it, and building a more inclusive sporting culture.

## **Outcomes**

### **1. A fair chance for every athlete**

More young athletes—especially from rural and low-income areas—will finally get equal opportunities to showcase their talent without needing expensive equipment or training centers.

### **2. Faster and more transparent talent selection**

Coaches and officials will receive accurate, AI-verified data that reduces human bias and speeds up the entire selection process.

### **3. Athletes improve through real-time insights**

Users will gain instant, easy-to-understand feedback on their performance, helping them practice better and track their progress over time.

### **4. A trusted and secure performance record**

Athlete results will be safely stored and verified, building confidence among players, parents, and sports authorities about the fairness of the system.

### **5. Lower costs for sports development programs**

Since assessments can be done with just a smartphone, organizations like schools, academies, and sports authorities will save money on equipment, staff, and manual evaluations.

### **6. Better decision-making for coaches and SAI officials**

With clear dashboards, leaderboards, and performance analytics, decision-makers can identify promising athletes earlier and support them more effectively.

### **7. Contribution to India's long-term sports growth**

By bridging the gap between technology and grassroots sports, the platform supports the nation's goal of nurturing world-class athletes from diverse backgrounds.

## **ABSTRACT**

**KhelQuest** is an AI-powered platform created to make sports talent assessment more accessible and fair for every aspiring athlete in India. Instead of relying on expensive equipment or specialized training centers, **KhelQuest** allows athletes to simply use their smartphones to record fitness tests. The app automatically analyzes movements using lightweight on-device AI models, evaluates performance, detects cheating, and provides instant feedback.

All results are securely stored in the cloud, where coaches and sports authorities can view verified dashboards and leaderboards, allowing them to discover promising talent from even the most remote regions. Designed to work smoothly on low-end devices and in areas with limited connectivity, the platform focuses on inclusivity, cost-effectiveness, and ease of use.

By combining smart technology with a mission of equal opportunity, **KhelQuest** aims to open doors for athletes across the country, ensure transparent evaluation, and support the nation's broader vision of identifying and nurturing sports talent.

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